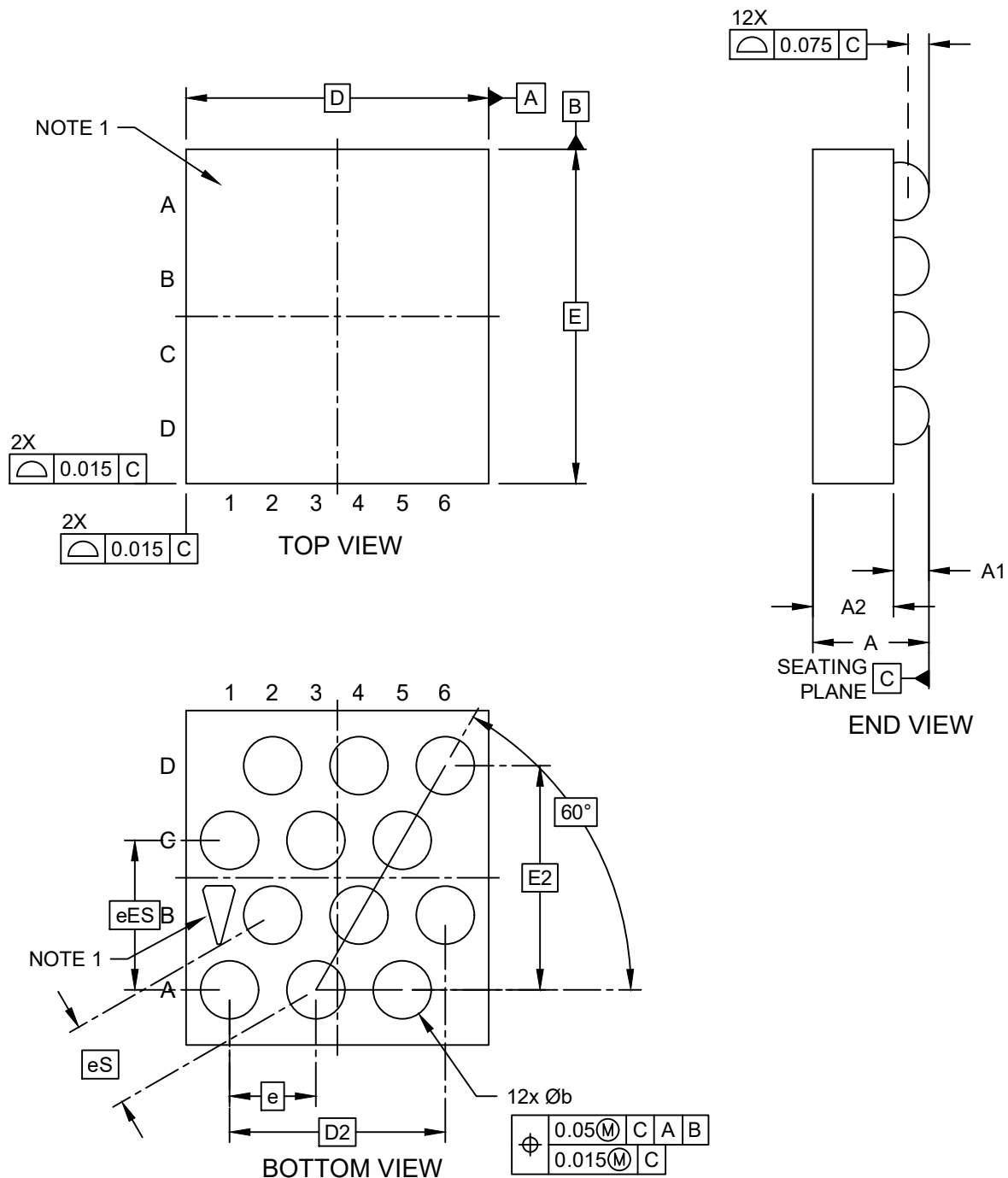


Packaging Diagrams and Parameters

12-Ball Wafer Level Chip Scale Package (F4B) - [WLCSP]

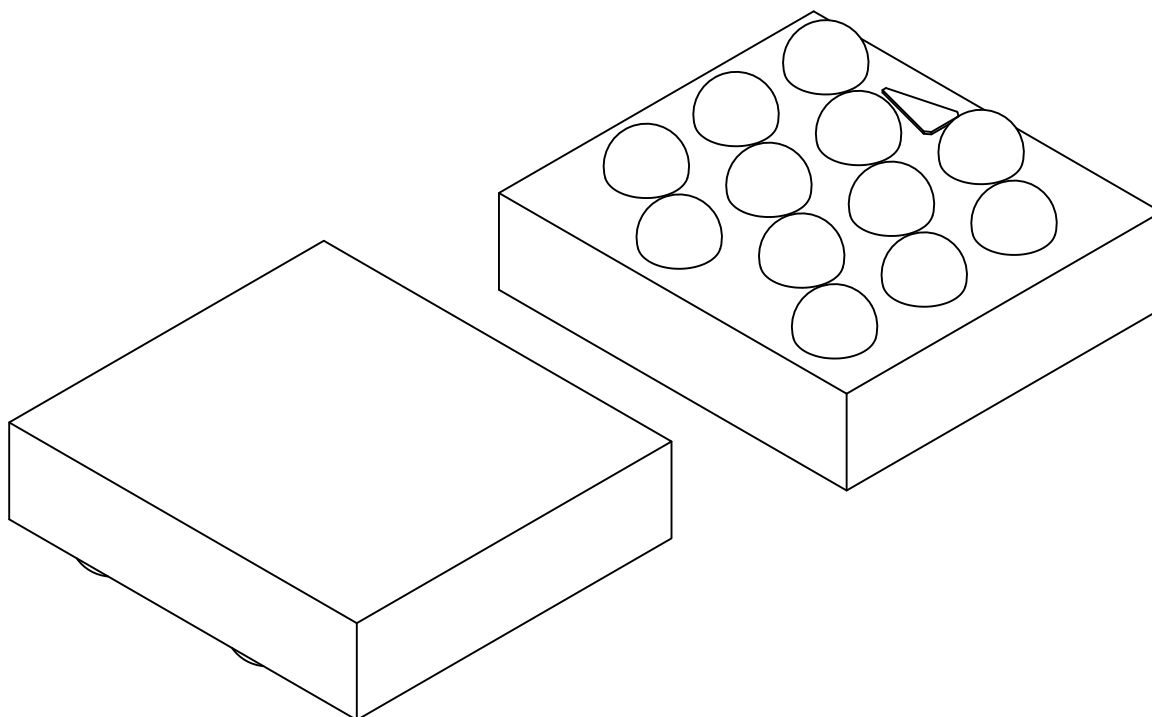
Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Packaging Diagrams and Parameters

12-Ball Wafer Level Chip Scale Package (F4B) - [WLCSP]

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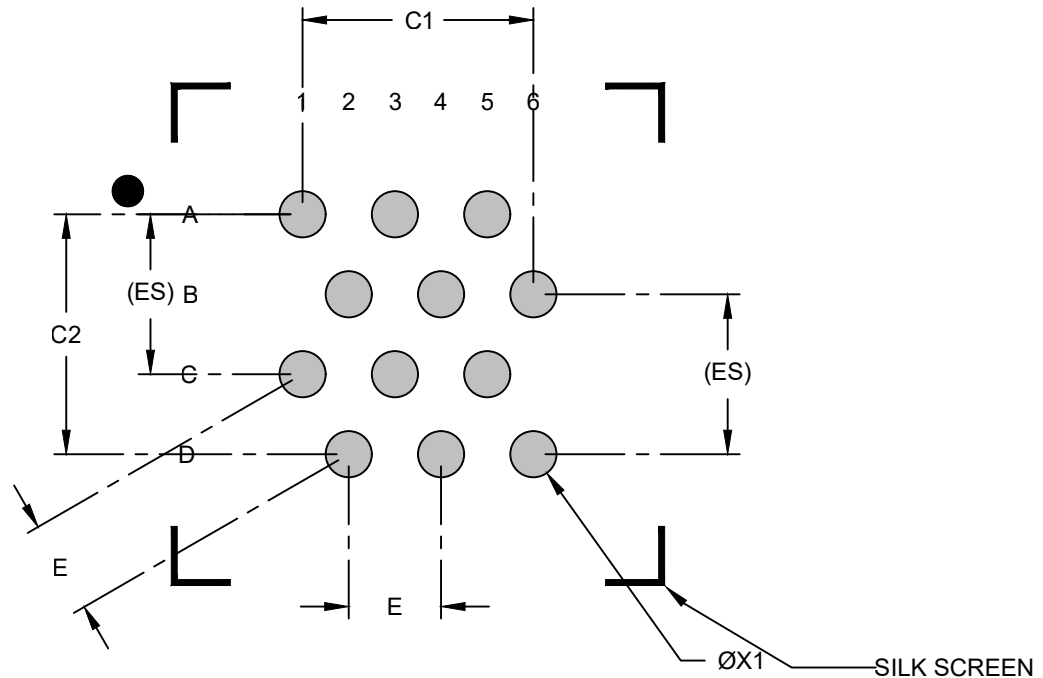
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	12		
Pitch	e	0.400 BSC		
Pitch	eS	0.400 BSC		
Pitch	eES	0.693 BSC		
Overall Height	A	–	–	0.538
Ball Height	A1	0.164	–	–
Die Thickness	A2	0.280	0.305	0.330
Overall Length (Max)	D	See Note 4.		
Ball Array Length	D2	1.000 BSC		
Overall Width (Max)	E	See Note 4.		
Ball Array Width	E2	1.039 BSC		
Ball Diameter	b	0.239	0.269	0.299

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.
- Please contact your local Microchip representative for details

12-Ball Wafer Level Chip Scale Package (F4B) - [WLSCP]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Ball Pitch	E	0.40 BSC		
Staggered Pitch	ES	0.693 REF		
Contact Pad Spacing	C1		1.000	
Contact Pad Spacing	C2		1.039	
Contact Pad Diameter (X12)	X1			0.200

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-23226 Rev A